

SOAP to REST Transformation Integration Flow

Business Scenario

A legacy ERP system communicates using SOAP-based web services, while the target cloud application exposes a REST/JSON API. SAP Integration Suite serves as the middleware layer, transforming the inbound SOAP request into a REST-compliant call, invoking the target API, processing the response, and returning a synchronous SOAP acknowledgment to the ERP system.

End-to-End Integration Flow

Step	Description
1	ERP initiates a SOAP request to SAP Integration Suite.
2	SOAP Sender Adapter validates the inbound request against the defined WSDL contract.
3	The SOAP envelope is removed, isolating the CustomerRequest payload.
4	Request Message Mapping transforms the CustomerRequest structure into the target REST payload format.
5	XML to JSON Converter converts the mapped XML payload into JSON.
6	Content Modifier sets the required HTTP headers, including Content-Type: application/json.
7	Request-Reply pattern invokes the HTTP Receiver Adapter.
8	HTTP Receiver Adapter calls the target REST API via an HTTP POST operation.
9	The REST API returns a JSON response.
10	JSON to XML Converter converts the JSON response back into XML.
11	Response Message Mapping transforms the RestResponse structure into CustomerResponse.
12	SOAP Sender Adapter wraps CustomerResponse within a SOAP envelope.
13	ERP receives the synchronous SOAP acknowledgment response.

Components Used

- SOAP Sender Adapter
- HTTP Receiver Adapter
- Request-Reply Pattern
- Content Modifier

- Message Mapping
- XML to JSON Converter
- JSON to XML Converter
- WSDL
- XSD
- Postman (for API testing)

Outcome

The end-to-end synchronous SOAP-to-REST integration was successfully implemented and validated using SAP Integration Suite, enabling seamless interoperability between the legacy ERP system and the target cloud application.